12 Java Interview Programs That Make You 60% Interview Ready

Objective: Write, explain and modify these programs to improve logic and interview confidence.

How to Use

- Understand the problem
- Write logic in steps
- Code yourself
- Dry run
- Explain in your own words

How to Think (Interview Approach)

- Understand input/output
- Break problem into steps
- Choose approach (loop, array, etc.)
- Write pseudocode
- Code cleanly
- Dry run with example

Theory You Must Know

- String vs StringBuilder
- Array basics
- Loops
- Basic time complexity

Common Mistakes

- Mugging code
- Not understanding logic
- Unable to explain
- No practice

1. Reverse String

Logic: Start from end and append characters

```
String str = "hello";
String rev = "";
for(int i = str.length()-1; i >= 0; i--){
   rev += str.charAt(i);
}
System.out.println(rev);
```

2. Palindrome Check

Logic: Reverse and compare

3. Factorial

Logic: Multiply 1 to n

4. Fibonacci

Logic: Sum of previous two numbers

5. Prime Number

Logic: Check divisibility till sqrt(n)

6. Swap Without Temp

Logic: Use + and -

7. Largest in Array

Logic: Compare each element

8. Vowels & Consonants

Logic: Check each character

9. Bubble Sort

Logic: Compare adjacent elements

10. Remove Duplicates

Logic: Use Set or loops

11. Star Pattern - Triangle

Logic: Nested loops

```
for(int i=1; i<=5; i++){
   for(int j=1; j<=i; j++){
      System.out.print("*");
   }
   System.out.println();
}
```

12. Star Pattern - Pyramid

Logic: Spaces + stars